GOOGLE CLOUD AI BUILDER CUP 2026 · RETAIL & COMMERCE

Taal

Gemini 3.5 Flash plans in an ADK agent. Deterministic guardrails price every rupee and hold out a control group on every play.

₹9,191 recovered vs doing nothing on one tea lot PROJECTED

The agent that sells what the forecast says you'll throw away.

TEAM

Nandish · lead

Rinu · co-builder

PROBLEM

1. Forecasts and offers live in different tools: one dark store writes stock off while the next runs out.
2. Customers get blanket markdowns, not offers matched to what they buy.
3. Nothing plans to the online sell-by cut-off: 30% of shelf life or 45 days at delivery.[1]

[1] FSSAI advisory to e-commerce FBOs, 3 Dec 2024, file RCD-13/1/2024-Regulatory-FSSAI(E-13150): minimum 30% shelf life or 45 days at delivery. Either-limit reading is Taal's configurable default.

THE PROBLEM · HOW BIG, WHICH RULE, WHO DECIDES

The forecast sees the write-off coming, but nothing acts on it.

HOW BIG · AS REPORTED

1.8%

of net order value (NOV) is lost to inventory at Blinkit.

As reported by Inc42, 22 Jul 2026. Per-store line below shows its inputs, from the same article.

₹17,132 cr ÷ 2,443 stores × 4 × 1.8% ≈ ₹0.5 cr per store a year

All inventory loss, one quarter annualised. Context for scale, not a Taal result.

THE RULE · AN ADVISORY, NOT LAW

30% or 45 days

of shelf life must remain at delivery of an online order.

FSSAI advisory to e-commerce food sellers, 3 Dec 2024. Stock can be in date on the shelf yet past its online sell-by, so the deadline can come weeks before expiry.

Either-limit reading is Taal's configurable default.

WHO IT IS FOR

A category or supply-chain head

They own write-off and are the buyer. Day to day, two people use Taal:

A demand planner checks the gap and the guardrails, then approves the play on the Play Desk.

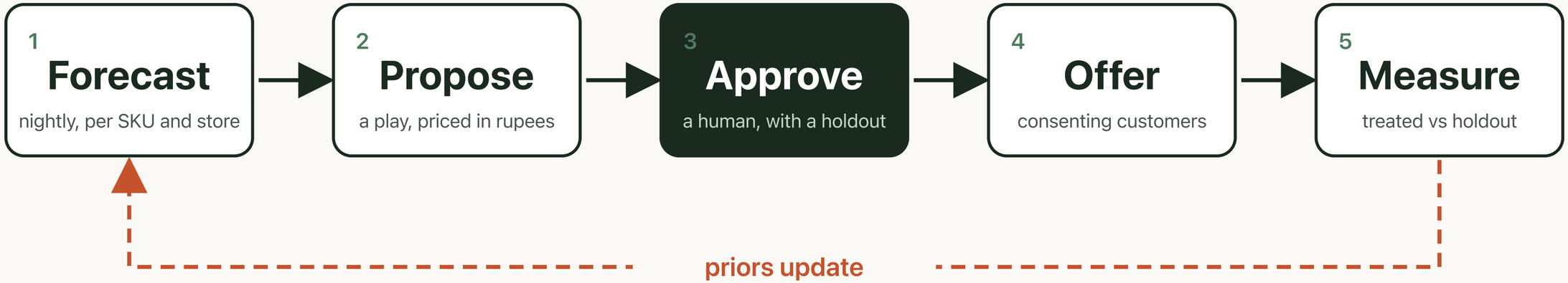
A dark-store manager confirms what is on the pallet on the phone view and receives any transfer.

NEXT The loop and the hero result are on slide 3; projected impact is on slide 15; 3 of 7 practitioners we asked write off weekly or daily (slide 13).

THE LOOP AND THE HERO PLAY

₹9,191 recovered vs doing nothing on one tea lot, ₹6,186 ahead of a blanket markdown

PROJECTED · SEEDED TENANT



TEA · GEMINI RUN · PROJECTED

₹9,191

recovered vs doing nothing (₹9,397 waste avoided – ₹206 transfer); ₹6,186 ahead of a blanket 20% markdown

HOLDOUT · SYNTHETIC ORDERS

+0.3 pp

pp = percentage points. 95% confidence interval (CI) −8.7 to +1.6. 1 of 355 treated responded, 0 of 39 held out. SYNTHETIC, inconclusive. Chips, 323 / 29 (19 Sep): also inconclusive

WHAT THIS PROVES

The loop closes

Measure runs end to end on synthetic orders; effect size needs a real pilot

THEME Taal forecasts stock per SKU and node nightly, with festival effects, and drafts human-approved offers.

DIFFERENTIATION

A holdout on every play. The play written back into the forecast.

	ONLINE SELL-BY CUT-OFF	HOLDOUT PER PROMO	PROMO → FORECAST WRITE-BACK	INDIA / Q-COMMERCE
Taal	Yes	Yes	Yes	Built for it; seeded tenant
Blue Yonder	Unclear	Unclear	Unclear	Partial
RELEX	Partial	Unclear	Partial	Partial
Wasteless	No	Unclear	Unclear	Unclear
Afresh	No	Unclear	Unclear	No
Flashfood	No	No	No	No

Unclear = no public evidence found in vendor pages read on 2026-10-09; it does not mean the feature is absent. Partial and No are our reading of those pages. Afresh and Flashfood are different product categories.

New: the model proposes, code owns every rupee, and every play keeps a holdout.

LIST OF FEATURES

Six capabilities, gated by rules a model never touches.

Vision intake

Photograph a pallet; dates and counts come back with confidence. Low confidence goes to a person.

Nightly Sense

Per-SKU, per-node forecasts. Six gap types, each with rupees and a deadline.

Planner Agent

ADK LoopAgent, max 3 iterations, six deterministic tools; a Cost Governor picks the gaps.

Approve + assignment

A human approves. Holdout by hash; offers go to treated customers only.

Customer Agent

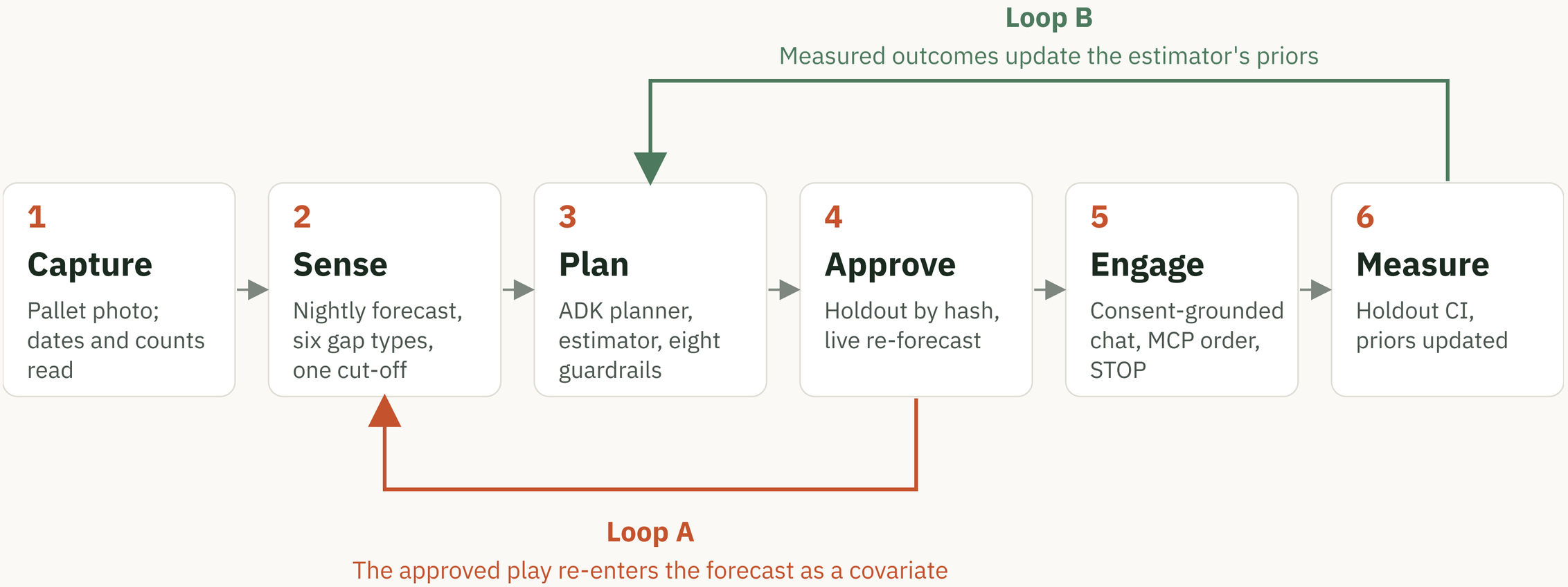
Grounded in stock, offers and consent. Orders through an in-process MCP tool.

Measure

Treated vs holdout, 95% CI. Priors update from the counts.

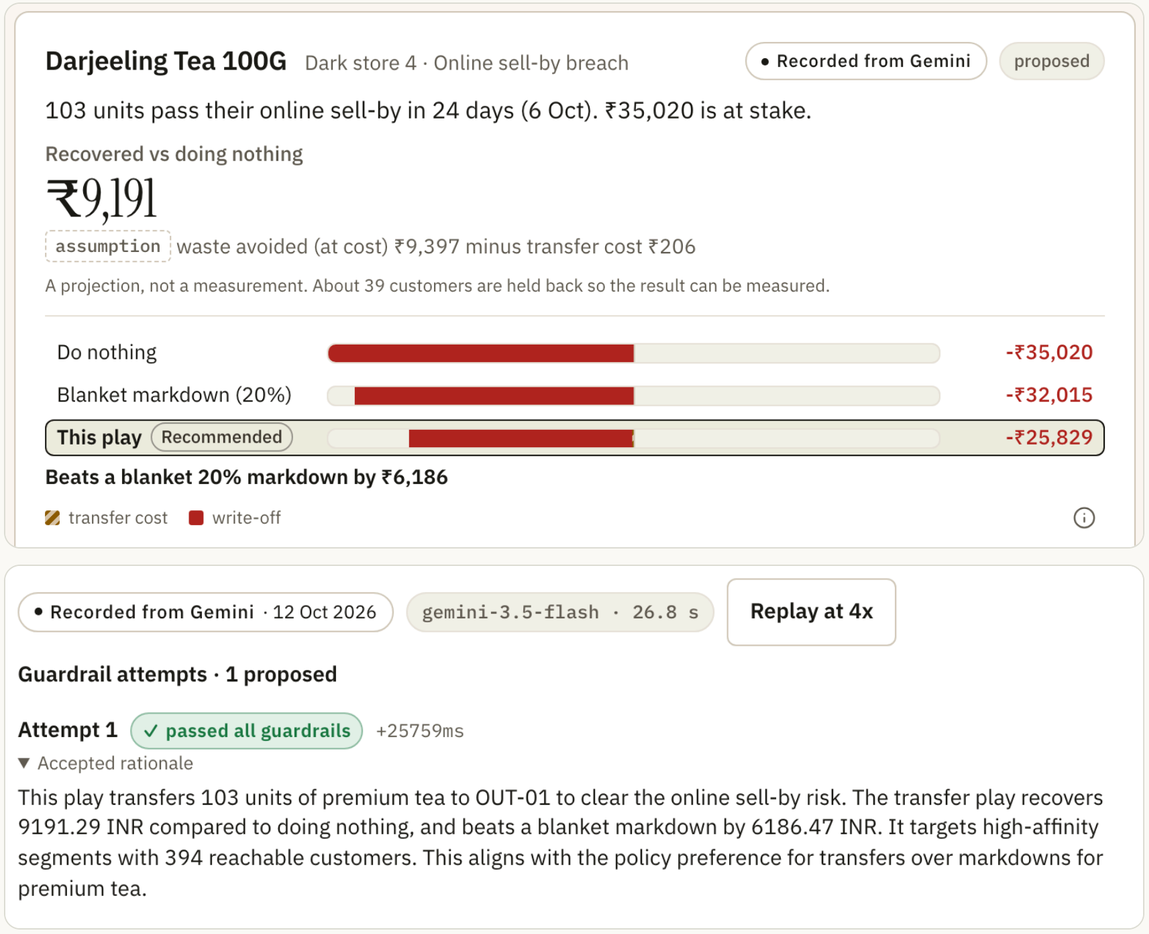
PROCESS FLOW

Two closed loops, not an open-ended pipeline.



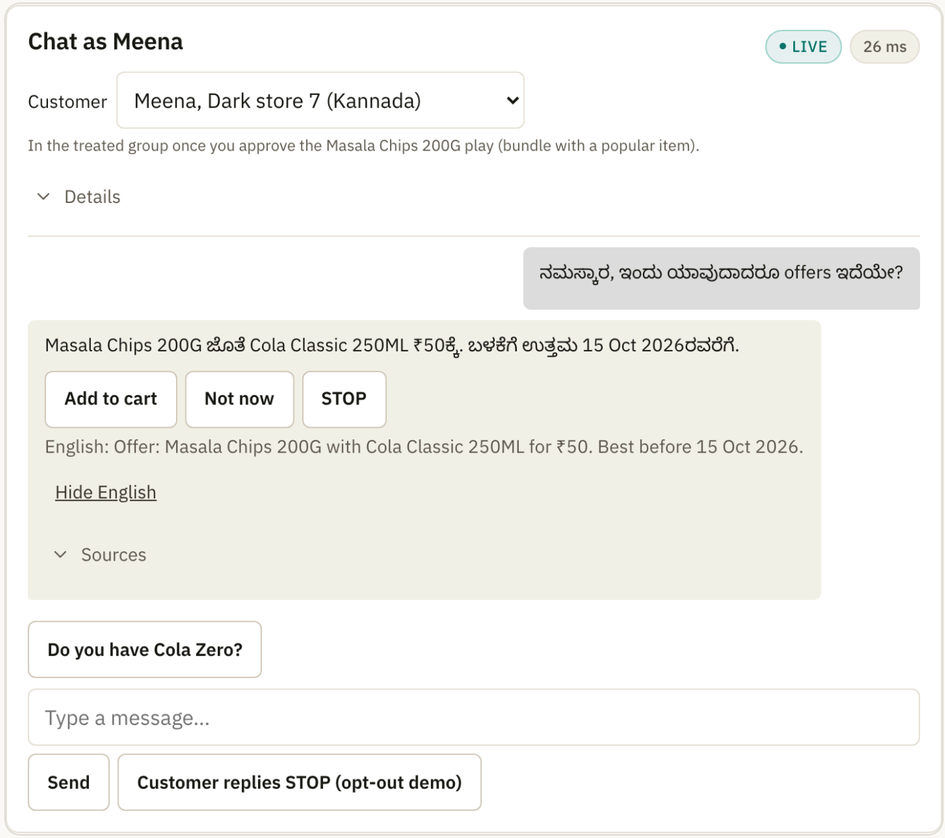
SCREENS

Gemini drafts the play; code prices every rupee; customers get Kannada plus English.



Play Desk: the tea play, recorded from Gemini, 12 Oct

Gemini drafted the play and its rationale; code priced every rupee and the guardrails passed it on the first attempt. Projected, seeded tenant.

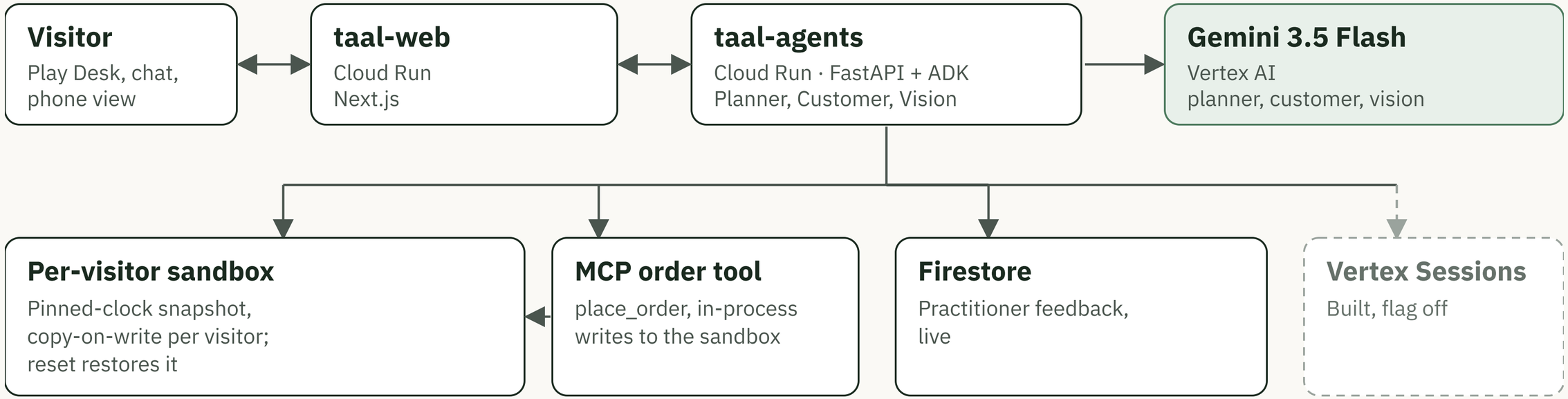


Customer chat, live

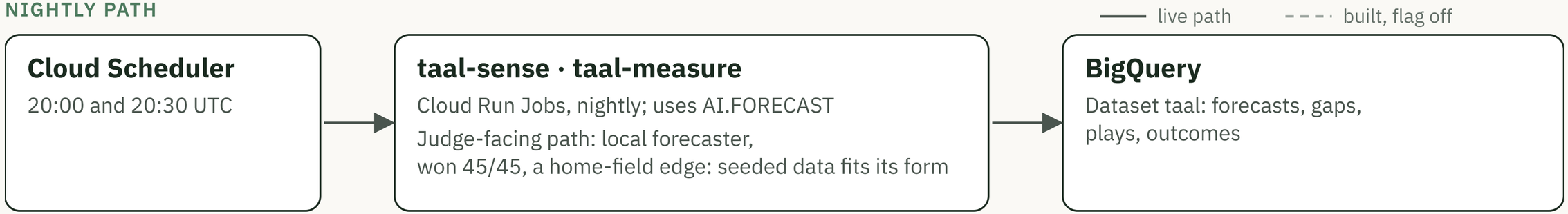
Kannada reply with an English gloss; the order goes through the MCP order tool.

ARCHITECTURE

A sandbox serves judges; nightly jobs run on BigQuery.



NIGHTLY PATH



TECHNOLOGIES USED

Each Google service, and the decision it makes.

SERVICE	THE DECISION IT MAKES
Google ADK planner (Gemini 3.5 Flash)	Proposes the play: mechanic, audience, window, in a LoopAgent (max 3) with 6 tools
Gemini on Vertex AI	Drafts the offer, chats with the customer, reads the pallet photo
Model Armor	Screens every customer chat message and reply for prompt injection and jailbreak (live since 14 Oct, revision taal-agents-00149-sw6)
BigQuery AI.GENERATE_TABLE	Writes the offer copy before Approve; used if ready within 3 s, else templated copy. The API reports which (copy.source).
Local forecaster (BigQuery AI.FORECAST nightly)	Re-forecasts the write-off on Approve; the nightly batch feeds Sense
Firestore	Stores practitioner feedback, live in the deployed service
Cloud Run, Jobs and Scheduler	Serves the app and API; runs Sense and Measure nightly; scales to zero
MCP order tool	Places the customer's order when they accept the offer
Deterministic guardrails and estimator	Check every play against 8 rules and own every rupee; Gemini never sets a number

DEPLOY GitHub Actions runs verify, then deploys to Cloud Run on every merge to main.

ESTIMATED IMPLEMENTATION COST

Priced from measured usage; sold on rupees rescued.

PER 30 DAYS, MODELLED FROM MEASURED USAGE

₹486 / $5.86

Pre-switch (gemini-2.5-flash, 23 Sep). Not re-measured on gemini-3.5-flash.

Measured with no plays in the window: idle-plus-nightly cost.

Real Gemini tokens, 752 BigQuery jobs and Cloud Run time, priced at public list rates. Not a bill.

Method: modeled_from_measured_usage.

Forward-looking profiles ESTIMATES

Profile	Scale	Monthly
Micro	1 shop, ≤500 SKUs	₹200–500
Mid	300 SKUs, 10 nodes	₹2,500–5,000
Large	20,000 SKUs, 500 nodes	₹1–2 lakh

PLANNER PER PLAY · SEPARATE FIGURE, MODELLED

about ₹6.5 per play

gemini-3.5-flash, modelled at list prices, 14 Oct ($0.0777 at ₹83 per $)

BUYER AND PRICING

Buyer: the category or supply-chain head, who owns write-off.
Basis: a fee per node per month, or a percentage of rupees rescued vs the holdout.

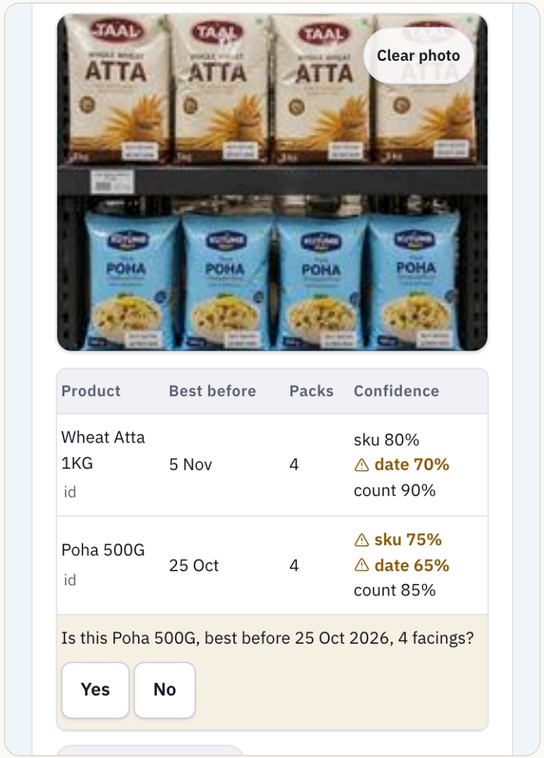
PROJECTED MARGIN LESS RUN COST ILLUSTRATION

≈ ₹21,710 – ₹156 to ₹313 per node

Per 28 days at the 5% prior: ₹347,356 margin over 16 nodes, less Mid-profile cost (₹2,500–5,000 a month, estimate). The ₹486 (pre-switch) had no plays in its window. Synthetic.

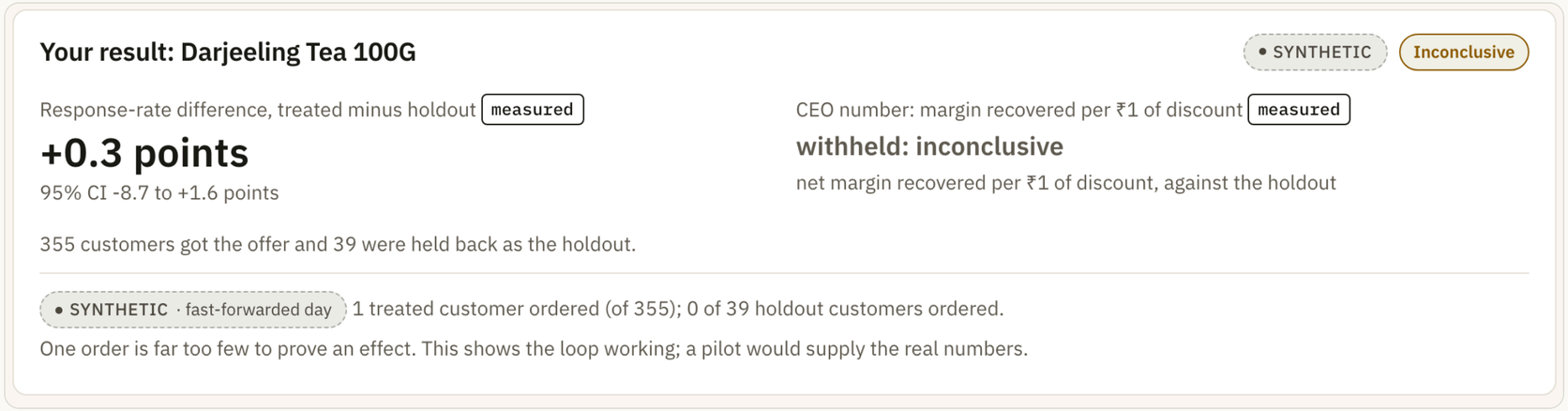
SNAPSHOTS OF THE PROTOTYPE

Doubtful reads go to a person; outcomes say SYNTHETIC.



Phone view

A read under 70% confidence is asked back before it reaches inventory.



Outcomes on a fast-forwarded day, synthetic orders: Tea

355 offered, 39 held out; 1 treated customer ordered, 0 held-out. Response rate +0.3 percentage points, 95% confidence interval –8.7 to +1.6. The range crosses zero, so the page says inconclusive and withholds the rupee figure.

PERFORMANCE REPORT

Measured and dated, on gemini-3.5-flash.

LIVE PLAN — 11 Oct

22/24

plans written by the model on the deployed app, 0 fallbacks

`plan_live_2026-10-12.json`

CUSTOMER CHAT — 14 Oct

3.66 s p50

p95 4.36 s, down from 8.44 s; Model Armor on

`customer_latency_2026-10-14_before_after`

APPROVE — 13 Oct

1.4–1.7 s

warm, five samples; 3.3 s before

`approve_timing_2026-10-13c.json`

PHOTO INTAKE — 14 Oct

30/30

dates read correctly on synthetic photos, not real phone photos

`vision_synthetic.json`

ADVERSARIAL EVAL — 14 Oct

50 of 60 attack runs stopped

0 of 18 genuine requests refused. 4 attack cases got through; 3 are now fixed in code, 1 photo case is not. Model Armor screens every chat message.

`eval/evaluation.md (adversarial)`

PLANNER EVALSETS — 14 Oct

39/46

first proposal accepted by the guardrails

`live_eval_2026-10-14/summary.json`

TESTS IN CI — every PR

960+

pytest and Playwright in make verify, before every deploy

`.github/workflows/verify.yml`

Full method, caveats and every source file: eval/evaluation.md and deck/claims.md. Forecast backtests: eval/external_backtest.md.

VALIDATION · PRACTITIONER RESPONSES, SEPTEMBER 2026

What seven practitioners told us.

3 of 7
write off stock weekly or daily

3 of 6
online sellers say the sell-by rule limits what they sell

0 of 7
measure a promotion against a control group

4 of 7
want a one-week pilot

"Blanket markdowns erode margin on products that may still sell normally."

Category / merchandising, quick-commerce

"Item-store recommendations with incremental-margin estimates."

Category / merchandising, quick-commerce

7 usable responses (10 collected, 3 excluded as flagged in config/feedback_exclusions.json). Anonymous by consent; contacts never published. Counts, not percentages.

USER GUIDE

Three people, three screens, one approved play.

Planner

/desk

1. Open the Play Desk and pick a gap.
2. Check the guardrails and counterfactuals.
3. Click Approve. The forecast moves.

Node manager

/phone

1. Photograph a pallet, or tap a sample.
2. Confirm the rows read below 0.7 confidence.
3. Approve the play from the gap card.

Customer

/chat

1. Open the chat. It starts with "Any offers today?".
2. Ask about a product. You see a stock band, not a count.
3. Send STOP to withdraw consent.

Badges. LIVE: a real call this visit. REPLAY: stored output replayed. SYNTHETIC: fabricated demo data. "Recorded from Gemini": a dated real model run.

IMPACT · WHAT IS MEASURED, WHAT IS PROJECTED

Projected margin hinges on one assumption.

TWO LOTS · PROJECTED

Tea (recorded Gemini run)

₹9,191 recovered

Vs doing nothing; projected, seeded tenant. ₹6,186 ahead of a blanket 20% markdown. 103 units, ₹35,020 at stake.

Chips (10 Oct, API snapshot)

-₹279

Projected. 368 units, ₹9,200 at stake. Net −₹8,468 vs −₹8,189.

RESPONSE PRIOR SENSITIVITY · PROJECTED

PRIOR	MARGIN	WASTE AVOIDED
2%	₹1.41 lakh	₹3.26 lakh
5% default	**₹3.47 lakh**	**₹7.06 lakh**
10%	₹6.19 lakh	₹11.18 lakh

Seeded tenant, 28-day horizon, 550 grocery gaps; plays planned at each prior: 421 (2%), 424 (5%), 423 (10%).

MEASURED VS PROJECTED

Measured on real customers: none yet.

Synthetic holdout, Tea: 355 treated, 39 held back; 1 responder vs 0, inconclusive.

Synthetic holdout, Chips (19 Sep): 323 treated, 29 held back; +1.5 points, 95% CI −12.4 to +5.6, crosses zero.

Projected at 5%: ₹347,356 margin, ₹706,183 waste avoided.

Practitioners: 7 usable responses; 3 of 7 write off stock weekly or daily.
Scale of the problem: sourced loss figure and arithmetic on slide 2 (as reported, all inventory loss, not a Taal result).

LONGEVITY · RUNNING IT FOR REAL

What it takes to run beyond the demo.

Production path

Today: services run as the owner-role deploy identity; CI uses a stored key. Secrets sit in a managed store.

Next: least-privilege service accounts and workload identity.

Model IDs pinned with retirement dates: gemini-3.5-flash live, Flash-Lite fallback; the previous model retires 20 Oct 2026.

Scalability

Measured: Sense 3.8 s (local, 300 SKUs); chat p50 3.66 s, p95 4.36 s (14 Oct, gemini-3.5-flash, Model Armor on, local API); live plan p50 26.5 s, p95 31.8 s (22/24, 11 Oct); Approve 1.4 to 1.7 s warm (13 Oct; first call after idle takes longer).

Estimate: 20,000 SKUs, 500 nodes. ESTIMATE

Breaks first: max-instances 1 with re-plan state in memory, then the planner queue.

Feasibility

Buyer: category or supply-chain head. Fee per node or rupee share.

Pilot: 50 to 60 testers, one week, not yet run.

Run cost: ₹486 a month, pre-switch (gemini-2.5-flash, 23 Sep), not a bill.

Estimates: ₹200 to 500 (Micro), ₹2,500 to 5,000 (Mid), ₹1 to 2 lakh (Large). ESTIMATE

ADDITIONAL DETAILS · FUTURE DEVELOPMENT

Next: a real pilot, real photos, more tenants.

1 A pre-registered pilot

One week, 50–60 consented testers, 40–50% holdout. Metric pre-registered in docs/pilot.md.

DESIGNED

2 Vision on real photos

Harness built. Real camera photos are the missing input; 30/30 (14 Oct) is on synthetic ones.

HARNESS BUILT

3 Multi-tenant BigQuery

Tenant id on every table; the BigQuery path is built, flag off until a second tenant.

BUILT, FLAG OFF

LEARNING CURVE · MEASURED, CHANGED, RE-MEASURED

MEASURED	BEFORE	WHAT WE CHANGED	AFTER
Chat p95, same 24 prompts	8.44 s	stock prefetch, Model Armor client reuse, minimal thinking	**4.36 s, Model Armor on (14 Oct)**
Approve, warm	3.3 s (12 Oct)	offer copy generated ahead of Approve	**1.4–1.7 s (13 Oct, five samples)**
Live planner	no_play, 110.9 s (20 Sep, pre-switch)	gap lookup moved into code	**22 of 24 model-authored (11 Oct)**
Planner evalsets	43/46 match, 1/46 trajectory (old recordings)	re-recorded on gemini-3.5-flash	**46/46, 18/46: regression guard, not accuracy**
Adversarial eval	10 of 60 attack runs got through	code-only fixes	**consent change refused by the tool; V3 3/3, P3 3/3; V6 not fixed**

Rows are copied from docs/learning_curve.md, which also lists the changes that made a number worse. Pre-switch means gemini-2.5-flash.

LINKS

Taal: the agent that sells what the forecast says you'll throw away

DEMO VIDEO

drive.google.com/file/d/1hBKapntHMnc4FNybRqSEWods2MTrluBw

GITHUB REPOSITORY

https://github.com/Nandish3010/probable-fortnight

LIVE APP

https://taal-web-2obkp776ca-el.a.run.app

₹9,191 recovered vs doing nothing on one tea lot · PROJECTED, seeded tenant

22 of 24 live plans on gemini-3.5-flash, p50 26.5 s (11 Oct)

Holdout so far: synthetic and inconclusive; a real pilot is next

Open the live app, click Run the 60-second beat, then Approve. Nothing you do there is saved.

Team Taal Google Cloud AI Builder Cup 2026 · Retail & Commerce